a specific developer. For example, to identify amino acids the strip is sprayed with 0.5% ninhydrin dissolved in acetone to visualize the colored spots indicating a particular amino acid. The ratio of the distance travelled by a component (i.e. amino acid) to that travelled by the solvent front, both measured from the marked point of application of the mixture, is called the resolution front or retardation factor (Rf) value for that component. Thus,

$$R_f = \frac{\text{Distance travelled by the solute}}{\text{Distance travelled by the solvent front}}$$

10.3.3 Adsorption chromatography

Adsorption is the process through which some substances physically bind to the surface of a solid substance like charcoal. Mobile phase is either a liquid (liquid – solid chromatography) or gas (gas-solid chromatography). Adsorption chromatography is based on the interaction between the solute molecules and active sites on the stationery phase. If stationary phase is much polar than the mobile phase, high polar compounds in the sample will bind tightly to the stationary phase while less polar compounds will bind less to the stationary phase. Therefore, the less lightly bound compound will be eluted out by the mobile phase earlier than the one that binds tightly.

10.3.4 Thin layer chromatography

Principle: Thin layer chromatography (TLC) is adsorption chromatography performed on adsorbent materials supported on glassplates. This technique combines the advantages of paper chromatography with those of column chromatography. In TLC, the stationery phase (alumina powder or silica gel with calcium sulphate as binder) is coated on a glass, plastic or foil plates. The thin layer is air dried at room temperature and activated in a hot air oven at 100 - 250°C. The mixture to be separated is spotted on the plate at one end and kept vertical in a container with mobile phase. After a few minutes, the components are separated through the thin layer and dried. The spots are detected by spraying the plates with specific staining reagents.

TLC is superior to paper Chromatography for the following reasons:

1. Concentrated sulfuric acid spray followed by heating may be used to develop the chromatogram by charring, which makes the spots of organic compounds visible.

2. The time taken to separate a mixture of amino acids by paper chromatography can be reduced by TLC.

3. The choice of adsorbents to allow separation of components is restricted in paper chromatography. In TLC, a wide range of adsorbents can be used.

Unit 10.indd 261 21-05-2018 15:13:12